MINI PROJECT
DOCUMENTATION

AI Plant Disease Detection

An AI-assisted web application concept for early plant disease awareness

Project Title	AI Plant Disease Detection
Development Tool	Base44 AI App Builder
Live Project	https://hungry-crop-guard-scan.base44.app
Project Type	Mini Project / AI-enabled web application prototype

Prepared for academic submission and GitHub project documentation.

Note: This document describes the project concept and a suggested workflow. Confirm the exact features and model behavior in the deployed application before presenting them as implemented.

1. Abstract

Plant diseases can reduce crop quality and yield, particularly when symptoms are not noticed early or expert advice is difficult to access. **AI Plant Disease Detection** is a web application concept intended to help users inspect possible plant-health problems by submitting a photograph of an affected leaf. The application can present a possible disease label, related symptom information, and practical next-step guidance when these functions are implemented and available in the deployed version.

The prototype is built using Base44, an AI app-building platform that can generate app interfaces, workflows, data handling, and deployment from natural-language instructions. The aim of this mini project is to demonstrate how an accessible web interface can support early plant-health awareness. The output should be treated as an initial screening aid, not as a guaranteed diagnosis.

2. Introduction

Agriculture is affected by fungal, bacterial, viral, and other plant-health problems. In many cases, early symptoms appear as spots, yellowing, curling, wilting, or unusual patterns on leaves. These signs may look similar across different diseases, so identifying the cause from appearance alone can be difficult. A simple digital tool can help users organize observations and find guidance before deciding on further action.

This project explores an AI-assisted website that makes the first step easier: upload a plant image, review the result, and learn what to check next. It is designed with farmers, home gardeners, students, and agriculture learners in mind.

3. Problem Statement

Farmers and plant growers may not have immediate access to an agricultural expert when they notice unusual symptoms on a crop. Delayed investigation can allow a disease or pest problem to spread and may lead to avoidable crop losses. The project addresses the need for an easy-to-access web interface that can assist with preliminary plant disease identification from leaf images and provide understandable follow-up information.

4. Objectives

- Provide a simple interface for submitting a plant or leaf image.
- Explore AI-assisted recognition of visible plant disease symptoms.
- Present results in clear, beginner-friendly language.
- Offer general prevention and next-step guidance where supported.
- Make the tool accessible through a web link on common devices.
- Demonstrate how an AI app builder can be used to create a practical prototype.

5. Scope of the Project

The project focuses on preliminary image-based plant-health support. Its practical scope depends on the features and integrations currently enabled in the published Base44 app. It is not intended to replace laboratory testing, field inspection, or advice from a qualified agricultural officer. Accuracy may vary with image quality, lighting, camera angle, crop variety, disease stage, and the range of diseases supported by the model.

6. Proposed System and Workflow

The intended user journey is shown below. Use this as the documented workflow only after checking that the live website supports each step.

1. Open website	2. Upload leaf image	3. Analyze image	4. Review result	5. Follow guidance
User visits the published app.	Choose a clear image of the affected plant.	The app processes the image if image-analysis functionality is configured.	A possible label or result is shown if supported.	Use guidance to decide whether expert confirmation is needed.

7. Main Features (Intended)

1. **Web-based access:** users can open the project through a browser.
2. **Image submission:** a user can provide a plant/leaf photograph if upload is enabled.
3. **AI-assisted analysis:** the application may classify an image when an appropriate model or analysis integration is connected.
4. **Result presentation:** show the likely issue and any confidence or explanatory details that the app actually provides.
5. **Plant-care guidance:** offer general prevention or treatment information if included.
6. **Responsive interface:** aim for usability on desktop and mobile screens.
7. **Expandable design:** the workflow can be improved with additional crops, languages, and expert resources.

Important: Features listed as intended are not a claim that every feature is currently implemented. Check the live app and edit this section to match what can actually be demonstrated.

8. Technology Used

Component	Role in the project
Base44 AI App Builder	Used to create and publish the web application from natural-language instructions; available capabilities depend on the project configuration.
Web browser	Used by users to access and interact with the published application.
Image input	Plant/leaf photograph supplied by the user, if the upload workflow is available.
AI/image analysis integration	A model or connected service is required for genuine image-based disease classification. Confirm the specific model/provider in the app configuration.
Hosting / deployment	The published Base44 project URL provides online access to the app.

9. System Requirements

Hardware: a computer or smartphone with a camera or access to an image file; internet connectivity.
Software: a modern browser such as Chrome, Edge, Firefox, or Safari; access to the published Base44 app.
Development: a Base44 account and the relevant project workspace for editing, testing, and publishing.

10. Implementation Methodology

The project can be developed in the following stages:

• **Stage 1 — Define requirements:** describe the target users, problem, inputs, expected results, and safety limitations.
• **Stage 2 — Generate the app:** use Base44 prompts to create the core interface and workflow.
• **Stage 3 — Configure analysis:** ensure a real image-analysis model or service is connected if disease classification is advertised.
• **Stage 4 — Add guidance:** provide clear, conservative plant-care information and encourage expert confirmation for serious cases.
• **Stage 5 — Test:** try supported images, unsupported files, poor lighting, unclear leaves, and mobile layouts.
• **Stage 6 — Publish and document:** publish the app, record its link, and maintain project documentation in GitHub.

11. Testing Plan

Test case	Expected behavior
Open the project URL	The website loads without an error.
Upload a clear leaf image	The app accepts the file and proceeds through the configured workflow.
Upload an unsupported file	The app rejects it or shows a helpful validation message, if validation is implemented.
Use a blurry or dark image	The app should ask for a clearer image or clearly communicate uncertainty where supported.
Check result details	Displayed disease names and guidance should be understandable and should not imply certainty without evidence.
Open on a mobile screen	Core content and controls remain readable and usable.

12. Advantages

• Easy to access through a web link.
• Can raise awareness of visible plant-health symptoms.
• May help users seek advice earlier.
• AI app-building tools can reduce the amount of manual setup needed for an initial prototype.
• The concept can be extended with regional languages, crop-specific information, and links to agricultural experts.

13. Limitations

• An image alone may not distinguish diseases with similar symptoms.
• The system may perform poorly on crops or diseases not represented in its model or data.
• Results depend on image quality and correct use.

- Internet access may be required.
- The exact AI model, dataset, and measured accuracy must be documented only if they are known and verified.
- Advice may not be suitable for every region, crop stage, or local regulation.

14. Responsible Use and Safety

Treat any automated result as a preliminary indication. Do not apply pesticides or other chemicals solely on the basis of an unverified AI result. Follow product labels and local agricultural guidance, and consult an agricultural officer or plant-health specialist when symptoms are severe, spreading, or unclear. If the app stores uploaded images or user details, explain what is collected and how it is handled.

15. Future Enhancements

• Add verified crop-specific disease classes and evaluate model performance on local field images.
• Add regional-language support for wider accessibility.
• Include offline-friendly educational material and low-bandwidth design.
• Provide location-aware seasonal advisories only with user consent and reliable data.
• Add a history page for previous scans if secure storage is needed.
• Connect users to agricultural extension services or expert review.
• Track model performance with a labelled test set and publish transparent evaluation metrics.

16. Conclusion

AI Plant Disease Detection demonstrates a practical application of AI-assisted web development for agriculture. By making plant-image submission and understandable follow-up guidance accessible through a website, the project aims to support early awareness of possible crop-health problems. Its real-world usefulness depends on the quality of the image-analysis model, the reliability of the guidance, and testing with representative plant images. Expert confirmation remains important before taking costly or potentially harmful action.

17. GitHub Repository Guide

Suggested repository structure:

AI-Plant-Disease-Detection/
■■■ README.md
■■■ docs/
■ ■■■ AI_Plant_Disease_Detection_Project_Documentation.pdf
■■■ screenshots/
■ ■■■ (add screenshots of the live app)
■■■ LICENSE (optional; choose a license appropriate to your work)

Recommended README contents

• Project title and short description
• Live demo link: https://hungry-crop-guard-scan.base44.app
• Problem statement and objectives
• Features verified in the deployed app
• Tool used: Base44 AI App Builder
• Screenshots and setup/access instructions
• Limitations, responsible-use note, and future improvements
• Link to this PDF in the docs folder

How to upload the PDF to GitHub

1. Create a new repository on GitHub, for example **AI-Plant-Disease-Detection**.
2. Open the repository and select **Add file** → **Upload files**.
3. Upload this PDF. For a tidy structure, place it inside a folder named **docs**.
4. Add a short commit message such as “Add mini project documentation PDF”.

5. Select **Commit changes**.
6. Add a link to the PDF in your README, for example: `[Project Documentation](docs/AI_Plant_Disease_Detection_Project_Documentation.pdf)`.

18. References

• Base44, “AI App Builder — Create an App for Free in Minutes.” https://base44.com/ai-app-builder
• Base44, “How to build an app with AI in 6 steps,” published August 12, 2026. https://base44.com/blog/how-to-build-an-app-with-ai
• Project deployment URL supplied for this documentation: https://hungry-crop-guard-scan.base44.app

Appendix: Project Details to Confirm Before Submission

Fill in or verify these details before your final academic submission:
• Student name(s), roll number(s), department, and institution.
• Guide/faculty name, academic year, and submission date.
• Exact features visible in the live application.
• Whether image analysis is connected to a real model; if so, name the model/provider and data source.
• Actual test cases and outcomes. Do not report accuracy percentages unless measured using a documented test dataset.